

PART A — MODULE 1 QUESTIONS

MATH5-M1-001

Prompt:

What is the solution to $5x - 12 = 2x + 9$?

- A) 3
- B) 5
- C) 7
- D) 21

MATH5-M1-002

Prompt:

A parking garage charges \$6 for the first hour and \$3 for each additional hour. What is the total charge for parking for 5 hours?

- A) \$15
- B) \$18
- C) \$21
- D) \$30

MATH5-M1-003

Prompt:

The table shows the numbers of pets owned by a group of students.

Number of pets: 0, 1, 2, 3

Number of students: 5, 8, 4, 3

How many students own at least 2 pets?

- A) 4
- B) 7
- C) 12
- D) 20

MATH5-M1-004

Prompt:

A linear function f has $f(0) = -3$ and $f(4) = 5$. What is $f(6)$?

- A) 7
- B) 8
- C) 9
- D) 11

MATH5-M1-005

Prompt:

Which inequality represents the solution to $3k + 8 \leq 29$?

- A) $k \leq 7$
- B) $k \geq 7$
- C) $k \leq 37$
- D) $k \geq 37$

MATH5-M1-006

Prompt:

A school club has 48 members. The number of juniors is 6 more than the number of seniors. How

many juniors are in the club?

- A) 21
- B) 24
- C) 27
- D) 30

MATH5-M1-007

Prompt:

Which expression is equivalent to $4x + 10$?

- A) $2(2x + 5)$
- B) $2(4x + 5)$
- C) $4(x + 10)$
- D) $10(x + 4)$

MATH5-M1-008

Prompt:

Which expression is equivalent to $(2x)^3$?

- A) $2x^3$
- B) $6x$
- C) $6x^3$
- D) $8x^3$

MATH5-M1-009

Prompt:

A standard number cube has faces numbered 1 through 6. If the cube is rolled once, what is the probability of rolling an odd number?

- A) $1/6$
- B) $1/3$
- C) $1/2$
- D) $2/3$

MATH5-M1-010

Prompt:

Two angles are complementary. One angle measures 34° . What is the measure, in degrees, of the other angle?

- A) 34
- B) 56
- C) 90
- D) 146

MATH5-M1-011

Prompt:

In the xy-plane, points P and Q have coordinates $(5, -1)$ and $(5, 6)$, respectively. What is the distance between P and Q?

- A) 5
- B) 6
- C) 7
- D) 11

MATH5-M1-012

Prompt:

If $h(x) = x^2 - 5$, what is $h(3)$?

- A) 1
- B) 4
- C) 6
- D) 14

MATH5-M1-013

Prompt:

The equation $(x + 3)(x - 2) = 0$ has two solutions. What is the smaller solution?

- A) -3
- B) -2
- C) 2
- D) 3

MATH5-M1-014

Prompt:

A rectangular poster has a length of 18 inches and a width of 12 inches. What is the perimeter, in inches, of the poster?

- A) 30
- B) 48
- C) 60
- D) 216

MATH5-M1-015

Prompt:

A pair of shoes costs \$72 before tax. If the sales tax is 5%, what is the total cost of the shoes, including tax?

- A) \$73.60
- B) \$75.00
- C) \$75.60
- D) \$77.00

MATH5-M1-016

Prompt:

The table shows the cost of several identical pens.

Number of pens: 2, 4, 6, 8

Cost in dollars: 3, 6, 9, 12

At this rate, what is the cost, in dollars, of 10 pens?

- A) 13
- B) 15
- C) 18
- D) 20

MATH5-M1-017

Prompt:

The formula $A = (1/2)(b_1 + b_2)h$ gives the area A of a trapezoid with bases b_1 and b_2 and height h . What is A when $b_1 = 6$, $b_2 = 10$, and $h = 5$?

- A) 21
- B) 40

- C) 50
- D) 80

MATH5-M1-018

Prompt:

Which value is equal to $\sqrt{64} + \sqrt[3]{8}$?

- A) 8
- B) 10
- C) 12
- D) 16

MATH5-M1-019

Prompt:

A line in the xy-plane is represented by $y = 7$. Which statement about the line is true?

- A) The line is vertical.
- B) The line is horizontal.
- C) The line has slope 7.
- D) The line passes through the origin.

MATH5-M1-020

Prompt:

A classroom has 18 desks arranged in equal rows. If there are 3 desks in each row, how many rows are there?

- A) 3
- B) 6
- C) 9
- D) 15

MATH5-M1-021

Prompt:

The expression $6s$ gives the total number of pencils in s identical boxes. What does the number 6 represent in this expression?

- A) The number of boxes
- B) The number of pencils in each box
- C) The total number of pencils
- D) The number of pencils left over

MATH5-M1-022

Prompt:

A data set contains the values 4, 6, 9, 11, and 15. Which value in the data set is closest to the mean?

- A) 6
- B) 9
- C) 11
- D) 15

PART B — MODULE 2 QUESTIONS

MATH5-M2-001

Prompt:

What is the solution to $2(5x + 3) = 4x + 42$?

- A) 4

- B) 5
- C) 6
- D) 8

MATH5-M2-002

Prompt:

The system of equations has solution (x, y) .

$$5x - y = 18$$

$$2x + y = 17$$

What is the value of $3x + y$?

- A) 19
- B) 22
- C) 25
- D) 28

MATH5-M2-003

Prompt:

The function f is defined by $f(x) = x^2 - 10x + 21$. What are the x -intercepts of the graph of $y = f(x)$?

- A) $(-7, 0)$ and $(-3, 0)$
- B) $(-3, 0)$ and $(7, 0)$
- C) $(3, 0)$ and $(7, 0)$
- D) $(0, 3)$ and $(0, 7)$

MATH5-M2-004

Prompt:

Which expression is equivalent to $(x^2 - 4x - 12)/(x - 6)$, where $x \neq 6$?

- A) $x - 6$
- B) $x - 2$
- C) $x + 2$
- D) $x + 6$

MATH5-M2-005

Prompt:

The function g is defined by $g(x) = -2(x - 3)^2 + 12$. What is the maximum value of $g(x)$?

- A) -12
- B) -3
- C) 3
- D) 12

MATH5-M2-006

Prompt:

A plant was 10 centimeters tall when it was first measured. Its height increased by 20% each week for 2 weeks. What was the plant's height, in centimeters, after 2 weeks?

- A) 12
- B) 14
- C) 14.4
- D) 15

MATH5-M2-007

Prompt:

The table shows values of a linear function f .

x : $-3, 0, 6, 9$
 $f(x)$: $14, 8, -4, -10$

Which equation defines f ?

- A) $f(x) = -2x + 8$
- B) $f(x) = -x + 11$
- C) $f(x) = 2x + 8$
- D) $f(x) = 3x - 4$

MATH5-M2-008

Prompt:

A rectangular storage container has a volume of 240 cubic feet. Its length is 10 feet, and its width is 6 feet. What is its height, in feet?

- A) 4
- B) 6
- C) 8
- D) 12

MATH5-M2-009

Prompt:

In a right triangle, the side opposite angle θ has length 9, and the hypotenuse has length 15. What is $\tan \theta$?

- A) $3/5$
- B) $3/4$
- C) $4/5$
- D) $5/4$

MATH5-M2-010

Prompt:

A line has equation $y = -(2/3)x + 4$. What is the x -intercept of the line?

- A) -6
- B) -4
- C) 4
- D) 6

MATH5-M2-011

Prompt:

The function f is defined by $f(x) = 4x + 1$. The function g is defined by $g(x) = x - 2$. What is $f(g(5))$?

- A) 9
- B) 13
- C) 15
- D) 21

MATH5-M2-012

Prompt:

What is the positive solution to $x^2 - 14x + 45 = 0$?

- A) 5
- B) 9
- C) 14
- D) 45

MATH5-M2-013

Prompt:

In the xy -plane, a circle has center $(2, 3)$ and passes through the point $(2, -1)$. Which equation represents the circle?

- A) $(x - 2)^2 + (y - 3)^2 = 4$
- B) $(x - 2)^2 + (y - 3)^2 = 16$
- C) $(x + 2)^2 + (y + 3)^2 = 4$
- D) $(x + 2)^2 + (y + 3)^2 = 16$

MATH5-M2-014

Prompt:

For what value of a does the equation $ax + 4 = 6x + 4$ have infinitely many solutions?

- A) 0
- B) 4
- C) 6
- D) 10

MATH5-M2-015

Prompt:

The table shows values of a function p .

x : $-2, -1, 0, 1$

$p(x)$: $9, 3, 1, 3$

Which equation could define p ?

- A) $p(x) = 2x^2 + 1$
- B) $p(x) = 2x^2 - 1$
- C) $p(x) = x^2 + 2x + 1$
- D) $p(x) = x^2 - 2x + 1$

MATH5-M2-016

Prompt:

Which expression is equivalent to $(5x^3y^4)(2x^2y)$, where x and y are positive?

- A) $7x^5y^5$
- B) $10x^5y^5$
- C) $10x^6y^4$
- D) $10x^6y^5$

MATH5-M2-017

Prompt:

A study group has 8 students. The mean number of practice questions completed by the students is 15. If 7 of the students completed a total of 98 practice questions, how many practice questions did the eighth student complete?

- A) 17
- B) 20
- C) 22
- D) 24

MATH5-M2-018

Prompt:

A rectangular prism has a square base with side length 5 centimeters and a height of 12 centimeters.

What is the surface area, in square centimeters, of the prism?

- A) 145
- B) 170
- C) 240
- D) 290

MATH5-M2-019

Prompt:

For all values of x , $x^2 + bx + 36 = (x + 9)(x + c)$, where b and c are constants. What is the value of b ?

- A) 4
- B) 9
- C) 13
- D) 36

MATH5-M2-020

Prompt:

The table shows the number of tickets sold for two performances.

Performance: Matinee, Evening

Adult tickets: 35, 50

Student tickets: 45, 30

For the two performances combined, what fraction of all tickets sold were adult tickets?

- A) $\frac{17}{32}$
- B) $\frac{8}{15}$
- C) $\frac{11}{20}$
- D) $\frac{5}{8}$

MATH5-M2-021

Prompt:

A store sells granola bars for \$1.25 each and bottles of water for \$1.75 each. A customer buys 12 items and spends \$18.00 before tax. How many granola bars did the customer buy?

- A) 4
- B) 5
- C) 6
- D) 8

MATH5-M2-022

Prompt:

The graph of a quadratic function has vertex $(4, -5)$ and passes through the point $(6, 3)$. Which equation could represent the function?

- A) $y = 2(x - 4)^2 - 5$
- B) $y = 2(x + 4)^2 - 5$
- C) $y = -2(x - 4)^2 - 5$
- D) $y = (x - 4)^2 + 3$

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH5-M1-001 — Correct: 7 — Explanation: Subtract $2x$ from both sides to get $3x - 12 = 9$. Add 12 to get $3x = 21$, so $x = 7$.

MATH5-M1-002 — Correct: \$18 — Explanation: The first hour costs \$6, and the 4 additional hours cost $4 \times \$3 = \12 . The total charge is $\$6 + \$12 = \$18$.

MATH5-M1-003 — Correct: 7 — Explanation: Students who own at least 2 pets are those who own 2 or 3 pets. There are $4 + 3 = 7$ such students.

MATH5-M1-004 — Correct: 9 — Explanation: The rate of change is $(5 - (-3))/(4 - 0) = 8/4 = 2$. Since $f(0) = -3$, $f(6) = 2(6) - 3 = 9$.

MATH5-M1-005 — Correct: $k \leq 7$ — Explanation: Subtract 8 from both sides to get $3k \leq 21$. Divide by 3 to get $k \leq 7$.

MATH5-M1-006 — Correct: 27 — Explanation: Let s be the number of seniors. Then juniors are $s + 6$, and $s + s + 6 = 48$. Thus $2s = 42$, so $s = 21$ and juniors = 27.

MATH5-M1-007 — Correct: $2(2x + 5)$ — Explanation: Factoring out 2 from $4x + 10$ gives $2(2x + 5)$.

MATH5-M1-008 — Correct: $8x^3$ — Explanation: $(2x)^3 = 2^3x^3 = 8x^3$.

MATH5-M1-009 — Correct: $1/2$ — Explanation: The odd numbers on a standard number cube are 1, 3, and 5, so the probability is $3/6 = 1/2$.

MATH5-M1-010 — Correct: 56° — Explanation: Complementary angles sum to 90° , so the other angle is $90 - 34 = 56^\circ$.

MATH5-M1-011 — Correct: 7 — Explanation: The points have the same x-coordinate, so the distance is the difference in y-values: $6 - (-1) = 7$.

MATH5-M1-012 — Correct: 4 — Explanation: Substitute 3 for x : $h(3) = 3^2 - 5 = 9 - 5 = 4$.

MATH5-M1-013 — Correct: -3 — Explanation: The equation is true when $x + 3 = 0$ or $x - 2 = 0$, so the solutions are -3 and 2. The smaller solution is -3.

MATH5-M1-014 — Correct: 60 — Explanation: The perimeter of a rectangle is $2(\text{length} + \text{width})$, so $2(18 + 12) = 60$.

MATH5-M1-015 — Correct: \$75.60 — Explanation: The tax is 5% of \$72, which is \$3.60. The total cost is $\$72 + \$3.60 = \$75.60$.

MATH5-M1-016 — Correct: 15 — Explanation: The cost per pen is $3 \div 2 = \$1.50$, so 10 pens cost $10 \times \$1.50 = \15 .

MATH5-M1-017 — Correct: 40 — Explanation: Substitute into the formula: $A = (1/2)(6 + 10)(5) = (1/2)(16)(5) = 40$.

MATH5-M1-018 — Correct: 10 — Explanation: $\sqrt{64} = 8$ and $\sqrt[3]{8} = 2$, so the sum is 10.

MATH5-M1-019 — Correct: The line is horizontal. — Explanation: The equation $y = 7$ means every point on the line has y-coordinate 7, so the line is horizontal.

MATH5-M1-020 — Correct: 6 — Explanation: Divide the total number of desks by the number of desks in each row: $18 \div 3 = 6$.

MATH5-M1-021 — Correct: The number of pencils in each box — Explanation: In $6s$, s is the number of boxes and 6 is the number of pencils per box.

MATH5-M1-022 — Correct: 9 — Explanation: The mean is $(4 + 6 + 9 + 11 + 15)/5 = 45/5 = 9$, so the value closest to the mean is 9.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH5-M2-001 — Correct: 6 — Explanation: Expand to get $10x + 6 = 4x + 42$. Then $6x = 36$, so $x = 6$.

MATH5-M2-002 — Correct: 22 — Explanation: Add the equations to get $7x = 35$, so $x = 5$. Then $2(5) + y = 17$, so $y = 7$. Thus $3x + y = 15 + 7 = 22$.

MATH5-M2-003 — Correct: (3, 0) and (7, 0) — Explanation: Factor $x^2 - 10x + 21$ as $(x - 3)(x - 7)$. The x-intercepts occur when $x = 3$ and $x = 7$.

MATH5-M2-004 — Correct: $x + 2$ — Explanation: Factor the numerator: $x^2 - 4x - 12 = (x - 6)(x + 2)$. Since $x \neq 6$, the expression simplifies to $x + 2$.

MATH5-M2-005 — Correct: 12 — Explanation: Since $g(x) = -2(x - 3)^2 + 12$ and the squared term

is always nonnegative, the maximum value occurs at $x = 3$ and is 12.

MATH5-M2-006 — Correct: 14.4 — Explanation: Increasing by 20% for 2 weeks means multiplying by 1.2 twice: $10(1.2)^2 = 14.4$.

MATH5-M2-007 — Correct: $f(x) = -2x + 8$ — Explanation: From $x = 0$ to $x = 6$, $f(x)$ changes from 8 to -4 , giving slope $-12/6 = -2$. Since $f(0) = 8$, $f(x) = -2x + 8$.

MATH5-M2-008 — Correct: 4 — Explanation: Volume equals length $\times$ width $\times$ height, so $240 = 10 \times 6 \times h$. Thus $h = 240/60 = 4$.

MATH5-M2-009 — Correct: $3/4$ — Explanation: The adjacent side is $\sqrt{(15^2 - 9^2)} = \sqrt{144} = 12$. Thus $\tan \theta = \text{opposite/adjacent} = 9/12 = 3/4$.

MATH5-M2-010 — Correct: 6 — Explanation: The x -intercept occurs when $y = 0$. Set $0 = -(2/3)x + 4$, so $(2/3)x = 4$ and $x = 6$.

MATH5-M2-011 — Correct: 13 — Explanation: First find $g(5) = 5 - 2 = 3$. Then $f(3) = 4(3) + 1 = 13$.

MATH5-M2-012 — Correct: 9 — Explanation: Factor $x^2 - 14x + 45$ as $(x - 5)(x - 9)$. The solutions are 5 and 9, so the positive solution among the choices intended is 9.

MATH5-M2-013 — Correct: $(x - 2)^2 + (y - 3)^2 = 16$ — Explanation: The radius is the distance from $(2, 3)$ to $(2, -1)$, which is 4. Thus $r^2 = 16$, and the equation is $(x - 2)^2 + (y - 3)^2 = 16$.

MATH5-M2-014 — Correct: 6 — Explanation: For the equation $ax + 4 = 6x + 4$ to be true for all x , the coefficients of x must be equal, so $a = 6$.

MATH5-M2-015 — Correct: $p(x) = 2x^2 + 1$ — Explanation: Substituting $x = -2, -1, 0$, and 1 into $2x^2 + 1$ gives 9, 3, 1, and 3, matching the table.

MATH5-M2-016 — Correct: $10x^5y^5$ — Explanation: Multiply coefficients and add exponents of like bases: $(5x^3y^4)(2x^2y) = 10x^5y^5$.

MATH5-M2-017 — Correct: 22 — Explanation: The total number of practice questions is $8 \times 15 = 120$. The eighth student completed $120 - 98 = 22$ questions.

MATH5-M2-018 — Correct: 290 — Explanation: The prism has two square bases with total area $2(5 \times 5) = 50$ and four rectangular faces with total area $4(5 \times 12) = 240$. The surface area is $50 + 240 = 290$.

MATH5-M2-019 — Correct: 13 — Explanation: Since the constant term is 36, $9c = 36$, so $c = 4$. Then $(x + 9)(x + 4) = x^2 + 13x + 36$, so $b = 13$.

MATH5-M2-020 — Correct: $17/32$ — Explanation: Adult tickets total $35 + 50 = 85$. All tickets total $35 + 50 + 45 + 30 = 160$. The fraction is $85/160 = 17/32$.

MATH5-M2-021 — Correct: 6 — Explanation: Let g be granola bars. Then bottles of water are $12 - g$. The cost equation is $1.25g + 1.75(12 - g) = 18$. This simplifies to $21 - 0.50g = 18$, so $g = 6$.

MATH5-M2-022 — Correct: $y = 2(x - 4)^2 - 5$ — Explanation: A quadratic with vertex $(4, -5)$ has form $y = a(x - 4)^2 - 5$. Using point $(6, 3)$, $3 = a(2^2) - 5$, so $8 = 4a$ and $a = 2$.